Chapter 4 Answer Key

Introduction to Statistics — MATH& 146

Section 4.2 Exercises

4.2.1. $X \sim N(100, 15)$. For each part, convert to z, then use the table.

(a) $P(X \leq 115)$: $z = \dfrac{115 - 100}{15} = \dfrac{15}{15} = 1.00$.

$$P(X \leq 115) = P(Z \leq 1.00) = \mathbf{0.8413}.$$

(b) $P(X > 85)$: $z = \dfrac{85 - 100}{15} = -1.00$.

$$P(X > 85) = P(Z > -1.00) = 1 - P(Z \leq -1.00) = 1 - 0.1587 = \mathbf{0.8413}.$$

Note: this is equal to (a) by symmetry — being more than 1 standard deviation above the mean and being more than 1 standard deviation below the mean are mirror-image events.

(c) $P(85 \leq X \leq 115) = P(Z \leq 1.00) - P(Z \leq -1.00) = 0.8413 - 0.1587 = \mathbf{0.6826}$.
This is precisely the Empirical Rule: $\approx 68\%$ of values lie within $\pm 1\sigma$ of the mean.

4.2.2. $X \sim N(20.1,\ 0.4)$ oz.

(a) $P(X < 20)$: $z = \dfrac{20 - 20.1}{0.4} = \dfrac{-0.1}{0.4} = -0.25$.
$P(Z \leq -0.25) = 1 - P(Z \leq 0.25) = 1 - 0.5987 = \mathbf{0.4013}$.
About 40.1% of boxes weigh less than 20 oz.

(b) $P(X < 19.5)$: $z = \dfrac{19.5 - 20.1}{0.4} = \dfrac{-0.6}{0.4} = -1.50$.
$P(Z \leq -1.50) = \mathbf{0.0668}$.
About 6.7% of boxes are underweight.

4.2.3. $X \sim N(50, 10)$. Find x such that $P(X \leq x) = 0.75$.

Step 1: Find z^* with $P(Z \leq z^*) = 0.75$. Searching the table: row 0.6, col 0.07 gives $P(Z \leq 0.67) = 0.7486$, which is the closest entry to 0.75.

Step 2: Convert back:

$$x = \mu + z^* \cdot \sigma = 50 + (0.67)(10) = 50 + 6.7 = \mathbf{56.7}.$$

Section 4.3 Exercises

4.3.1. Heights $\sim N(67, 3)$. The standard error $\sigma_{\bar{x}} = 3/\sqrt{n}$:

n	$\sigma_{\bar{x}} = 3/\sqrt{n}$
4	$3/2 = 1.500$
16	$3/4 = 0.750$
25	$3/5 = 0.600$
100	$3/10 = 0.300$

Pattern: Each time n is multiplied by 4, $\sigma_{\bar{x}}$ is cut in half (divided by 2). More precisely, $\sigma_{\bar{x}}$ is proportional to $1/\sqrt{n}$: quadrupling the sample size only halves the standard error. Larger samples give more precise estimates, but with diminishing returns.

4.3.2. $\sigma = 20$. We need $\sigma_{\bar{x}} = 20/\sqrt{n} < 2$.

$$\frac{20}{\sqrt{n}} < 2 \implies \sqrt{n} > 10 \implies n > 100.$$

The minimum sample size is $\mathbf{n = 101}$.

4.3.3. When we average n independently drawn values, the high observations and low observations tend to cancel each other out. A single measurement can land far from μ purely by chance; but for the average of many measurements to land far from μ, most of those measurements would have to fall on the same side of μ simultaneously — an increasingly unlikely coincidence as n grows. Mathematically, this cancellation is captured by the $1/\sqrt{n}$ factor in the standard error: the spread of $\bar{x}$ around μ shrinks as n increases, so $\bar{x}$ estimates μ more reliably.

Section 4.5 Exercises

4.5.1. Battery lifetimes: $\mu = 500$ hr, $\sigma = 40$ hr, $n = 64$.

(a) **Sampling distribution of $\bar{x}$:**

$$\bar{x} \sim N\left(500,\ \frac{40}{\sqrt{64}}\right) = N(500,\ 5).$$

Mean $= \mu_{\bar{x}} = 500$ hr; standard error $= \sigma_{\bar{x}} = 5$ hr.

(b) $P(\bar{x} \le 495)$:

$$z = \frac{495 - 500}{5} = -1.00. \qquad P(Z \le -1.00) = \mathbf{0.1587}.$$

(c) $P(496 \le \bar{x} \le 504)$:

$$z_1 = \frac{496 - 500}{5} = -0.80, \qquad z_2 = \frac{504 - 500}{5} = 0.80.$$

$$P(-0.80 \le Z \le 0.80) = P(Z \le 0.80) - P(Z \le -0.80) = 0.7881 - 0.2119 = \mathbf{0.5762}.$$

4.5.2. $\mu = 100$, $\sigma = 25$. Increasing from $n = 25$ to $n = 100$:

- $\mu_{\bar{x}}$ is unchanged: the center of the sampling distribution remains at $\mu_{\bar{x}} = 100$ regardless of n.
- $\sigma_{\bar{x}}$ decreases: $\sigma_{\bar{x}}$ goes from $25/\sqrt{25} = 5$ to $25/\sqrt{100} = 2.5$. The standard error is cut in half because the sample size quadrupled. The sampling distribution is narrower and more tightly centered around the true mean.

4.5.3. $P(\bar{x} > 535)$ is smaller than $P(X > 535)$ because sample means are far less variable than individual observations. For a single SAT score from $N(530, 115)$, the standard deviation is 115 — a score of 535 is only $5/115 \approx 0.04$ standard deviations above the mean, so it is very common. For the sample mean of $n = 50$ students, the standard error is $115/\sqrt{50} \approx 16.3$ — a sample mean of 535 is about $5/16.3 \approx 0.31$ standard errors above the population mean, still not extreme, but already further into the tail than the individual score appears. For large n, the sampling distribution clusters so tightly around 530 that even small departures from 530 become surprising.

More broadly: using sample means to make decisions is more powerful than using individual observations because sample means are far more predictable. A single outlier can make an individual observation look unusual without indicating anything about the population; a sample mean of 50 students that is unusually high is much harder to explain away as coincidence.

Chapter 4 Review

1. $X \sim N(7.5,\ 1.1)$ lbs (birth weights).

(a) $P(X < 6)$:

$$z = \frac{6 - 7.5}{1.1} = \frac{-1.5}{1.1} \approx -1.36.$$

$P(Z \le -1.36) = 1 - P(Z \le 1.36) = 1 - 0.9131 = \mathbf{0.0869}$.
About 8.7% of newborns weigh less than 6 pounds.

(b) $P(7 \le X \le 9)$:

$$z_1 = \frac{7 - 7.5}{1.1} \approx -0.45, \qquad z_2 = \frac{9 - 7.5}{1.1} \approx 1.36.$$

$P(Z \le -0.45) = 1 - P(Z \le 0.45) = 1 - 0.6736 = 0.3264$.

$$P(7 \le X \le 9) = 0.9131 - 0.3264 = \mathbf{0.5867}.$$

About 58.7% of newborns weigh between 7 and 9 pounds.

(c) Top 5%: $P(X > x) = 0.05 \Rightarrow P(Z \le z^*) = 0.95$. From the table, $z^* = 1.645$ (midpoint of $z = 1.64 \to 0.9495$ and $z = 1.65 \to 0.9505$).

$$x = 7.5 + 1.645(1.1) = 7.5 + 1.810 \approx \mathbf{9.31} \text{ lbs.}$$

(d) For $n = 36$:

$$\bar{x} \sim N\left(7.5,\ \frac{1.1}{\sqrt{36}}\right) = N\left(7.5,\ \frac{1.1}{6}\right) \approx N(7.5,\ 0.1833).$$

Mean = 7.5 lbs; standard error = $1.1/6 \approx 0.1833$ lbs.

(e) $P(\bar{x} > 7.8)$ for $n = 36$:

$$z = \frac{7.8 - 7.5}{1.1/6} = \frac{0.3}{0.1833} \approx 1.64.$$

$P(Z > 1.64) = 1 - 0.9495 = \mathbf{0.0505}$.

Only about 5% of samples of 36 newborns would have a mean birth weight above 7.8 lbs.

2. $\mu = 20$, $\sigma = 8$, $n = 40$. **Can we use the normal distribution?**

Yes. Although the population distribution is skewed, the sample size is $n = 40 \geq 30$, so the Central Limit Theorem guarantees that $\bar{x}$ is approximately normally distributed:

$$\bar{x} \approx N\left(20,\ \frac{8}{\sqrt{40}}\right) = N(20,\ 1.265).$$

Computing $P(\bar{x} \leq 19)$:

$$z = \frac{19 - 20}{8/\sqrt{40}} = \frac{-1}{1.265} \approx -0.79.$$

$P(Z \leq -0.79) = 1 - P(Z \leq 0.79) = 1 - 0.7852 = \mathbf{0.2148}$.

About 21.5% of samples of size 40 from this population will have a mean at or below 19.

3. The student is **incorrect.** The CLT applies to *sample means*, not to individual data values. A sample of 100 people drawn from a skewed population will still contain skewed data — the individuals themselves do not become normally distributed just because the sample is large. What the CLT guarantees is that the *distribution of the sample mean* $\bar{x}$ (computed across many different samples of size 100) is approximately normal. The raw data in any single sample retain the shape of the original population.

4. Model response (writing prompt):

Descriptive statistics in Chapter 1 produced $\bar{x}$ and s to summarize a sample; the z-score $z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$ turns that sample mean into a standardized distance from the claimed population mean, measured in units of the standard error. The denominator is $\sigma/\sqrt{n}$ rather than σ because we are asking how far the *average* of n observations strays from μ, and the CLT tells us that averages have standard deviation $\sigma/\sqrt{n}$, which is smaller than the individual standard deviation σ by a factor of $\sqrt{n}$. In Chapter 5, this z-score becomes the test statistic: we compare it to the standard normal distribution to assess how surprising the observed $\bar{x}$ would be if the null hypothesis about μ were true, completing the bridge from description to inference.